Fifty–fifty is not naive: equal allocation is the minimax design for treatment-effect estimation under unknown variances

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Abstract

Neyman's allocation rule minimizes the variance of the difference-in-means estimator of an average treatment effect by splitting the sample in proportion to the outcome standard deviations — but it needs those standard deviations, which is exactly what an experiment is run to learn. The default fallback, an even 50/50 split, is usually described as a concession to ignorance. We argue it is the opposite: **equal allocation is the unique minimax-optimal design when the variance ratio is unknown.** Concretely, against an adversary who chooses the two outcome variances, the even split guarantees relative efficiency 1/2 versus the oracle Neyman design, and no allocation guarantees more. We prove this with an exact worst-case identity, $\inf_t \rho(p,t) = \min(p, 1-p)$, give the sharp factor-of-two bound it specializes to, and then ask the practical question: a two-stage design can estimate the variances from a pilot and reallocate — when is it worth it? Simulation shows the answer is governed by a clean **efficiency ceiling** $\rho(\frac{f}{2} + (1-f)p^*,\, t)$ set by the equally-split pilot: adaptivity buys a lot when the variances are genuinely unequal, and is a small net *loss* when they are not.

1. Setup and notation

Two arms, control ($a = 0$) and treatment ($a = 1$), with outcomes of variance σ_0^2 and σ_1^2. A design allocates a fraction $1-p$ of the n units to control and p to treatment ($p \in (0,1)$), so $n_0 = (1-p)n$ and $n_1 = pn$. The average treatment effect $\tau = \mu_1 - \mu_0$ is estimated by the difference in sample means $\hat{\tau} = \bar{Y}_1 - \bar{Y}_0$, which is unbiased for any fixed design, with variance

$$V(p) \;=\; \frac{\sigma_0^2}{n(1-p)} + \frac{\sigma_1^2}{np}. \tag{1}$$

Minimizing (1) over p gives the **Neyman allocation** [1]

$$p^* \;=\; \frac{\sigma_1}{\sigma_0 + \sigma_1}, \qquad V(p^*) \;=\; \frac{(\sigma_0 + \sigma_1)^2}{n}. \tag{2}$$

Only the variance *ratio* matters, so normalize $\sigma_0 = 1$ and write $t = \sigma_1/\sigma_0 \in (0, \infty)$. Dropping the common $1/n$, let $\bar{V}(p;t) = \frac{1}{1-p} + \frac{t^2}{p}$, so that $\bar{V}(p^*) = (1+t)^2$ and $p^*(t) = t/(1+t)$. The **relative efficiency** of design p against the oracle that knows t is

$$\rho(p,t) \;=\; \frac{\bar{V}(p^*;t)}{\bar{V}(p;t)} \;=\; \frac{(1+t)^2}{\dfrac{1}{1-p} + \dfrac{t^2}{p}} \;\in (0,1]. \tag{3}$$

$\rho(p,t) = 1$ iff $p = p^*(t)$. The catch is that p^* depends on t, which is unknown before the experiment. The question of this note is what to do about that.

2. Results

2.1 The factor of two

A folklore fact, included because it sets the stage and is sharp.

Proposition 1. *For every $t > 0$,*

$$\frac{V(\frac{1}{2})}{V(p^*)} \;=\; \frac{2(\sigma_0^2+\sigma_1^2)}{(\sigma_0+\sigma_1)^2} \;=\; \frac{2(1+t^2)}{(1+t)^2} \;\in [1,2),$$

equal to 1 iff $t = 1$ and approaching 2 as $t \to 0$ or $t \to \infty$. Equivalently, equal allocation never wastes more than a factor of two in variance, and an even split of $2n$ units dominates the Neyman split of n units for every (σ_0, σ_1).

Proof. Substitute $p = \frac{1}{2}$ into (1)–(2): $V(\frac{1}{2}) = 2(\sigma_0^2+\sigma_1^2)/n$ and $V(p^*) = (\sigma_0+\sigma_1)^2/n$. The ratio is $2(1+t^2)/(1+t)^2$; since $2(1+t^2)-(1+t)^2 = (1-t)^2 \geq 0$ it is ≥ 1, with equality iff $t = 1$, and the limits as $t \to 0, \infty$ are 2. The supremum 2 is not attained on $(0,\infty)$. □

So in the *worst* case equal allocation costs a doubling of variance. But "worst case" here is over the variances t, and that reframing is the point: a design is a decision made *before* t is known, so the honest way to compare designs is by their guarantee across t.

2.2 Equal allocation is minimax

Treat the variance ratio t as chosen adversarially. The guaranteed (worst-case) efficiency of a design p is $g(p) = \inf_{t>0} \rho(p,t)$, and a *minimax* (maximin) design maximizes it. The next identity computes g exactly.

Theorem. *For every $p \in (0,1)$,*

$$\inf_{t>0} \rho(p,t) \;=\; \min(p,\, 1-p). \tag{4}$$

Consequently

$$\sup_{p\in(0,1)} \inf_{t>0} \rho(p,t) \;=\; \tfrac{1}{2},$$

attained uniquely at $p = \frac{1}{2}$. Equal allocation guarantees $\rho(\frac{1}{2},t) > \frac{1}{2}$ for every finite t — the infimum $\frac{1}{2}$ is approached only as the variance ratio degenerates ($t \to 0$ or ∞) — and no other design guarantees as much.

Proof. Relabeling the two arms sends $(p,\sigma_0,\sigma_1) \mapsto (1-p,\sigma_1,\sigma_0)$, i.e. $t \mapsto 1/t$, and leaves V and the Neyman optimum invariant; hence $\rho(p,t) = \rho(1-p,1/t)$. It therefore suffices to treat $p \leq \frac{1}{2}$, where $\min(p,1-p) = p$, and prove $\inf_t \rho(p,t) = p$.

Lower bound $\rho(p,t) \geq p$. From (3), $\rho(p,t) \geq p$ is equivalent to $(1+t)^2 \geq p(\frac{1}{1-p} + \frac{t^2}{p}) = \frac{p}{1-p} + t^2$, i.e.

$$1 + 2t \;\geq\; \frac{p}{1-p}. \tag{5}$$

For $p \le \frac{1}{2}$ we have $p/(1-p) \le 1 \le 1+2t$ for all $t \ge 0$, so (5) holds; thus $\rho(p,t) \ge p$ for all t.

Tightness. Taking $t \to \infty$ in (3), $\rho(p,t) = \dfrac{(1+t)^2}{\frac{1}{1-p} + \frac{t^2}{p}} \to p$. Hence the infimum equals $p = \min(p, 1-p)$, establishing (4). The supremum of $\min(p,1-p)$ over $(0,1)$ is $\frac{1}{2}$, attained only at $p = \frac{1}{2}$; for any $p \ne \frac{1}{2}$, $\min(p,1-p) < \frac{1}{2}$, so the even split is the unique maximizer.

Finally, at $p = \frac{1}{2}$, $\bar{V}(\frac{1}{2}; t) = 2(1+t^2)$ and $\rho(\frac{1}{2}, t) = \dfrac{(1+t)^2}{2(1+t^2)}$, which exceeds $\frac{1}{2}$ for all $t \ne 1$ because $(1+t)^2 - (1+t^2) = (1-t)^2 > 0$, equals 1 at $t=1$, and tends to $\frac{1}{2}$ as $t \to 0, \infty$ without attaining it. $\square$

Two remarks. First, (4) is strikingly simple: the worst case an adversary can inflict on design p is to put almost all the variance on the *under-allocated* arm, dragging efficiency down to the smaller of the two allocation fractions. Second, the theorem is a statement about *guarantees*, not averages — it does not say the even split is best on any particular instance (it is beaten by Neyman on every instance with $t \ne 1$), but that it is the safest bet when you must commit to a design without knowing t.

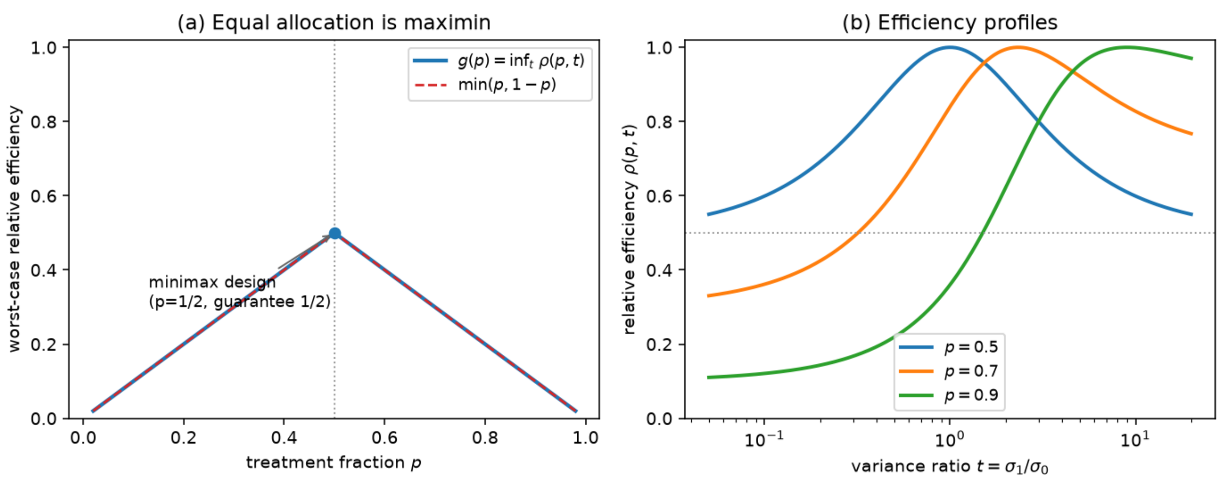

Figure 1: **Figure 1.** (a) The worst-case efficiency $g(p) = \inf_t \rho(p,t)$ (solid) coincides with $\min(p, 1-p)$ (dashed) to within grid resolution ($\max|\cdot| = 10^{-4}$), peaking at $(1/2, 1/2)$: equal allocation is the maximin design. (b) Efficiency profiles $\rho(p,t)$ vs. the variance ratio t for three designs; each peaks at its own t (where $p = p^*$) and decays away from it. The $p = 1/2$ curve is the flattest and the only one that never drops below 1/2.

Figure 1(a) confirms (4) numerically: over a fine grid the computed worst-case efficiency matches $\min(p, 1-p)$ to 10^{-4}, and the maximin is $p = 0.500$ with guarantee 0.500. A Monte-Carlo check (40{,}000 replications, $n = 2000$, $t = 4$) reproduces (1): empirical variances 0.0168 (equal) and 0.0125 (Neyman) match the analytic 0.0170 and 0.0125, with ratio 1.36 — well inside the factor-of-two envelope.

2.3 So should you adapt? A two-stage design and its ceiling

The minimax result is a worst-case insurance argument. In practice one often has *some* data: run a pilot, estimate the variances, and reallocate. Does that recover the oracle? We study the natural

two-stage design: spend a fraction f of the budget on an equally-split pilot, form $\hat{\sigma}_0, \hat{\sigma}_1$ from it, set $\hat{p} = \hat{\sigma}_1/(\hat{\sigma}_0 + \hat{\sigma}_1)$, and allocate the remaining $(1-f)n$ units by $\hat{p}$. All n outcomes (pilot included) enter $\hat{\tau}$.

There is a structural limit that has nothing to do with estimation noise. Even with a *perfect* pilot ($\hat{p} = p^*$), the equally-split pilot pins down part of the allocation, so the realized treatment fraction converges to

$$p_f(t) \;=\; \tfrac{f}{2} + (1-f)\,p^*(t), \tag{6}$$

a convex pull of p^* toward $\frac{1}{2}$. The design's first-order efficiency is therefore capped at

$$\rho(p_f(t),\, t) \;<\; 1 \quad (\text{unless } f = 0 \text{ or } t = 1). \tag{7}$$

This is the price of insisting on a balanced pilot: you commit a chunk of the sample to 50/50 before you know better, and that chunk can never be Neyman-optimal.

The simulation ($R = 20{,}000$ replications per cell, pilot fraction $f = 0.3$, seed `20260615`) bears this out precisely (Figure 2).

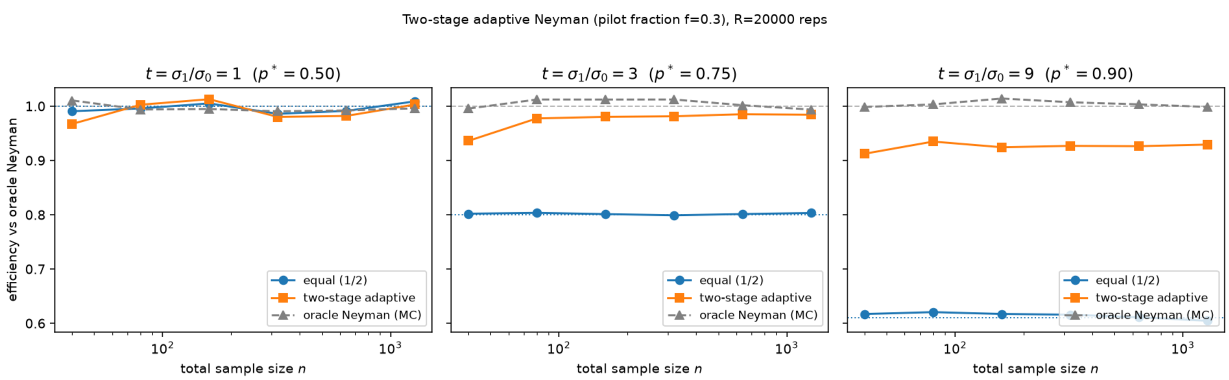

Figure 2: **Figure 2.** Efficiency relative to the oracle Neyman design vs. sample size n, for equal allocation (blue), the two-stage adaptive design (orange), and an oracle-Neyman Monte-Carlo control (grey). Dotted line: the analytic equal-allocation efficiency $\rho(1/2, t)$. **Left ($t = 1$):** the variances are equal, so 50/50 is already optimal and adaptivity is a small net loss at small n — it pays estimation noise to move away from the right answer. **Middle ($t = 3$):** equal sits at 0.80; adaptive climbs to its ceiling $\rho(0.675, 3) = 0.975$. **Right ($t = 9$):** equal sits at 0.61; adaptive recovers most of the gap but plateaus at its ceiling $\rho(0.78, 9) = 0.923$, *not* at 1.

The measured plateaus match (7) almost exactly: for $t = 3$, $p_f = 0.675$ and $\rho(0.675, 3) = 0.975$ against a measured ≈ 0.98; for $t = 9$, $p_f = 0.78$ and $\rho(0.78, 9) = 0.923$ against a measured ≈ 0.92–0.93. The equal-allocation curves sit flat at the predicted $\rho(\frac{1}{2}, 3) = 0.80$ and $\rho(\frac{1}{2}, 9) = 0.61$. And at $t = 1$ the adaptive design is a (small) net loss — efficiency 0.967 at $n = 40$, versus ≈ 1.0 for the even split — because there is nothing to gain and estimation noise in $\hat{p}$ only perturbs an already-optimal allocation.

The upshot for practice. Adaptivity is worth it exactly when the variances are genuinely unequal *and* you cannot guess the ratio in advance; the gain is bounded by (7), so a smaller pilot f raises the ceiling but makes $\hat{p}$ noisier. When the variances are comparable — the common case — the even split is not a fallback but the right answer, and chasing a Neyman correction can only cost you.

3. Discussion

The reframing is the contribution: the even split, usually justified by appeals to simplicity or symmetry, has an exact decision-theoretic optimality. Equation (4) makes the worst-case efficiency of *any* allocation a one-line formula, and the even split is its unique maximizer with the clean guarantee $\rho > 1/2$. This complements, rather than competes with, Neyman allocation: Neyman is the right target when you can credibly bound the variance ratio (and a pilot or prior gives you one), while 50/50 is what you are entitled to assume when you cannot. The two-stage ceiling (7) then quantifies how much of the Neyman gain a cheap pilot can actually deliver.

Relation to prior work. Neyman allocation dates to stratified sampling [1] and is standard in trial design [2]. The factor-of-two (Proposition 1) is folklore in the sampling literature [2]. The minimax framing of *equal allocation for ATE-estimation efficiency* (the Theorem) is, to our reading, not the one usually stated: the well-known minimax-optimality results for the even-then-Neyman scheme concern **best-arm identification** and *simple regret* [3,4], a different objective (selecting the larger mean, not estimating the contrast with minimal variance), and the **adaptive Neyman** literature [5,6] focuses on matching the oracle variance asymptotically, on small-pilot performance, and on the inferential subtleties of estimating the allocation online. Our point is orthogonal and finite-t: among all *fixed* designs, the even split is the exact maximin for estimation efficiency, with worst-case value $1/2$. The two-stage ceiling (6)–(7) is an elementary but, we think, underappreciated diagnostic for when adaptation pays.

Limitations. (i) The criterion is the variance of the difference-in-means estimator under independent arms with known (or consistently estimated) variances; we do not treat covariate adjustment, clustering, or non-Gaussian tail effects on the variance *estimate*. (ii) The minimax adversary ranges over the *ratio* $t \in (0, \infty)$ with no constraint; any credible bound $t \in [1/c, c]$ shrinks the worst case (the guarantee becomes $\rho(\frac{1}{2}, \cdot)$ evaluated at the endpoints, strictly above $1/2$) and can justify a mild departure from 50/50. (iii) The two-stage analysis reports the *unconditional* variance of $\hat{\tau}$ via simulation; valid *inference* after data-dependent allocation needs care, since the realized n_1 is random — the difference in means stays unbiased (assignment is independent of potential outcomes by design), but the usual plug-in variance estimator and its normal calibration deserve a separate treatment, as the adaptive-Neyman literature emphasizes [5,6]. (iv) We fixed the pilot fraction $f = 0.3$; jointly optimizing f (and going fully sequential) trades the ceiling (7) against estimation noise and is left open.

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